

UniMate AI

AI-Powered Smart University Platform

Project Documentation

Document No.: 04

Software Requirements Specification (SRS)

Part V

System Constraints, Assumptions & Dependencies

IEEE 29148 Inspired

Version: 1.0

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Academic Year

2025–2026

1. Introduction

This section defines the limitations, assumptions, and external dependencies that affect the design, development, operation, and future expansion of the UniMate AI platform.

Understanding these factors helps the development team make realistic decisions and ensures that project goals remain achievable within the available resources and timeline.

2. System Constraints

System constraints represent restrictions that influence the implementation and operation of UniMate AI.

2.1 Technical Constraints

CON-001: Platform Constraints

Description:

The mobile application will initially target Android devices only.

Requirements:

- Minimum Android Version: Android 10+
- Development Environment: Android Studio
- Programming Language: Kotlin

Impact:

iOS users will not be supported in Version 1.0.

Priority: High

CON-002: Backend Technology Constraint

Description:

The backend system must be developed using ASP.NET Core Web API.

Reason:

- Strong performance.
- Scalable architecture.
- Good support for REST APIs.
- Suitable for enterprise applications.

Impact:

The development team must follow .NET ecosystem standards.

Priority: High

CON-003: Database Constraint

Description:

The system shall use PostgreSQL as the primary database.

Reason:

- Reliability.
- Advanced relational features.
- Strong data consistency.

Impact:

Database design must follow PostgreSQL capabilities.

Priority: High

CON-004: AI Service Constraint

Description:

AI features depend on the availability and limitations of Gemini API.

Possible limitations:

- API quotas.
- Response limitations.
- Internet dependency.

Impact:

AI features may require optimization and request management.

Priority: High

2.2 Development Constraints

CON-005: Project Timeline Constraint

Description:

The system must be completed within the graduation project schedule.

Impact:

- Features must be prioritized.
- Development will follow MVP methodology.

Priority: High

CON-006: Team Resource Constraint

Description:

Development resources are limited.

Therefore:

- Features must be carefully selected.
- Complexity must be controlled.

Priority: Medium

CON-007: Budget Constraint

Description:

The project should primarily use free or educational service plans.

Examples:

- Firebase Free Tier.
- Supabase Free Tier.
- Gemini Free Tier.

Impact:

Resource usage must be monitored.

Priority: Medium

2.3 Security Constraints

CON-008: Data Protection Constraint

Description:

The system must protect user information and academic data.

Requirements:

- Secure authentication.
- Authorized access.
- Encrypted communication.

Priority: High

CON-009: Privacy Constraint

Description:

The system must avoid exposing sensitive user information.

Examples:

- Passwords.
- Authentication tokens.
- Private files.

Priority: High

3. System Assumptions

Assumptions represent conditions expected to be true during system operation.

ASM-001: Internet Availability

Description:

Users are expected to have an active internet connection.

Reason: The system depends on:

- Backend APIs.
- Cloud Storage.
- AI Services.

ASM-002: User Device Availability

Users are assumed to have compatible Android devices or modern browsers.

ASM-003: External Service Availability

The project assumes continuous availability of:

- Firebase Authentication.
- Supabase Storage.
- Gemini API.

ASM-004: User Technical Knowledge

Users are expected to have basic knowledge of:

- Mobile applications.
- Web interfaces.
- Digital systems.

ASM-005: Data Accuracy

The system assumes that users provide valid academic information.

Examples:

- Course details.
- Assignments.
- User profiles.

4. External Dependencies

External dependencies are services or technologies required for system operation.

DEP-001: Firebase Authentication

Purpose:

- User registration.
- Login.
- Password management.

Dependency Risk: Service availability and API changes.

DEP-002: Gemini API

Purpose:

- AI Assistant.
- Summarization.
- Academic support.

Dependency Risk:

- API limits.
- Service availability.

DEP-003: Supabase Storage

Purpose:

- File storage.
- Document management.

Dependency Risk:

- Storage limits.
- Network availability.

DEP-004: PostgreSQL Database

Purpose: Persistent data storage.

Dependency Risk:

- Database availability.
- Backup requirements.

DEP-005: GitHub Repository

Purpose:

- Source code management.
- Version control.
- Collaboration.

Dependency Risk:

- Repository access.
- Version management.

5. Business Constraints

BUS-001: Academic Project Scope

The project must satisfy graduation project requirements.

Therefore:

- Documentation is mandatory.
- Implementation must be demonstrable.
- Features must provide academic value.

BUS-002: MVP Priority

The first version focuses on essential features.

Priority Order:

- Authentication
- User Management
- Courses
- Files
- AI Assistant
- Assignments
- Dashboard

BUS-003: Future Expansion

Advanced features are planned for future versions.

Examples:

- AI Quiz Generator.
- Smart Study Planner.
- Attendance System.
- Learning Analytics.

6. Requirement Risks

Risk	Description	Mitigation
AI Service Limitation	Gemini API restrictions	Optimize requests
Time Limitation	Graduation deadline	Prioritize MVP
Integration Issues	Multiple technologies	Incremental development
Security Problems	Data exposure	Apply security standards
Storage Limits	Cloud restrictions	Monitor usage

7. Project Limitations

Version 1.0 has the following limitations:

LIM-001: Platform Limitation

Only Android is supported initially.

LIM-002: AI Accuracy Limitation

AI responses may require user verification.

LIM-003: Internet Dependency

Most features require an internet connection.

LIM-004: Limited University Integration

The system does not directly connect with existing university ERP systems.

LIM-005: Limited Automation

Advanced automation features are reserved for future versions.

8. Future Constraint Resolution

Future releases may reduce current limitations through:

- iOS application development.
- Local AI models.
- University system integration.
- Advanced cloud infrastructure.
- Automated workflows.

9. Summary Table

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Business Constraints	3
Limitations	5

10. Conclusion

The constraints, assumptions, and dependencies defined in this section provide a realistic foundation for developing UniMate AI. By identifying limitations early, the development team can make better technical decisions, manage risks effectively, and ensure successful delivery of Version 1.0.

End of Part V

Document No. 04 – Software Requirements Specification (SRS) – Part V: System Constraints, Assumptions & Dependencies
Version 1.0